# 5-Bar SCARA v2 Onboarding Package

M5

Encoder + Sensor Fusion Edition | Engineering Roadmap

**CALIBRATION & TEST LEAD**

## 1 | ROLE & OBJECTIVE

As **Calibration & Test Lead (M5)**, you are the quality gate of the entire project. Nothing proceeds to the next phase without your documented sign-off. Your responsibilities span the full project lifecycle: you are the **Reviewer for Phase 2B** (encoder closed-loop) and **Phase 3B** (Kalman sensor fusion), and you **co-own Phase 6** (robustness hardening and the 30-cycle unattended acceptance test).

**Primary Mission:** Design and execute rigorous verification protocols for each phase gate, harden the full system against real-world failure modes (lighting variation, stall events, power transients, Kalman divergence), and witness and certify the final 30-cycle acceptance run.

## 2 | HARDWARE & COMPONENTS REQUIRED

| Component | Purpose | Notes |
|---|---|---|
| Dial Indicator (0.01mm resolution) | Phase 1 runout check; position accuracy measurement | Magnetic stand for hands-free setup |
| Oscilloscope / Logic Analyzer | Phase 1-D encoder signal quality; stall timing verification | Verify sub-1ms trigger response |
| Calibration Dot Grid (3×3) | Phase 3/3B centroid accuracy and RMS error measurement | Print flat on rigid surface |
| Reference Test Objects (colored) | Known-position targets for centroid and pick accuracy tests | 3–5 objects of different colors |
| Stopwatch / millis() Serial logging | Latency measurements: vision pipeline, correction loop | Arduino Serial Plotter preferred |
| Power Supply with current meter | Phase 6: monitor current during 30-cycle run for relay transients | Log power events |
| 1000µF Electrolytic Capacitor | Phase 6 HW hardening: across DRV8825 VMot+/GND rail | Install per M5 power spec |
| 100nF Ceramic Capacitors | Phase 6 HW hardening: DRV8825 VCC and encoder signal decoupling | 4+ per roadmap |
| Ferrite Bead (optional) | Phase 5/6: relay coil flyback suppression if encoder drift observed | On relay coil line |

## 3 | SOFTWARE & PLATFORMS REQUIRED

| Tool | Purpose | Source |
|---|---|---|
| Python + numpy + pandas | Statistical analysis of test data: stddev, RMS, success rate | pip install |
| matplotlib / seaborn | Plot centroid noise, Kalman convergence, cycle time distribution | pip install |
| pyserial | Capture Serial output from both ESP32 nodes for automated logging | pip install pyserial |
| Arduino Serial Plotter | Real-time visualization of encoder counts and Kalman state | Built-in Arduino IDE |
| Spreadsheet (Excel / Sheets) | Track test results: PASS/FAIL per test ID, measured vs. target | Office / Google |
| Git | Access firmware branches; tag validated versions (e.g. v2B-PASSED) | git-scm.com |
| Test Log Template (create yours) | Standardized form: Test ID, date, tester, measured value, PASS/FAIL | Create from scratch |

## 4 | AI INTEGRATION STRATEGY

Your role is analytical by nature. Use AI to generate test scripts, analyze statistical results, identify failure root causes, and design the 30-cycle acceptance protocol:

### Prompt 1 — Generate automated test script for TEST 2B-A:

*"Write a Python script using pyserial to automate TEST 2B-A for an ESP32 encoder system. The script should: send a Serial command to move Motor 1 by +90 degrees, read the Serial response containing g_enc1_count, verify it equals ENC_CPR/4 (=600 counts), repeat 10 times, and report PASS if all within ±5 counts, FAIL otherwise. Include error handling for Serial timeout and connection loss."*

### Prompt 2 — Analyze Kalman filter test results:

*"I ran TEST 3B-A: 50 frames on a stationary object. Raw centroid X measurements (mm): [paste your 50 values]. Fused Kalman X output: [paste 50 values]. Calculate: (1) raw stddev, (2) fused stddev, (3) improvement ratio, (4) does the filter meet the <1mm fused stddev spec? If not, suggest which Kalman parameter (Q_POS, Q_VEL, R_MEAS) to adjust and by how much."*

### Prompt 3 — Design 30-cycle acceptance test protocol:

*"Design a detailed test protocol for a 30-cycle unattended robotic pick-and-place run. Include: pre-run checklist, cycle timing log format, success/failure criteria per cycle, how to classify stall events vs. missed picks vs. drop errors, what data to log at each cycle, and the pass/fail final scoring rubric matching: PASS 1: >=27/30 successful cycles; PASS 2: zero unrecoverable faults; PASS 3: zero undetected stalls."*

### Prompt 4 — Power hardening analysis:

*"My SCARA robot uses a DRV8825 stepper driver, an electromagnet relay, and encoders all on the same power rail. During relay switching I see occasional encoder count drift (+/-3 counts). Explain the electrical cause and provide the exact capacitor and ferrite bead placement to eliminate this. What is the flyback voltage spike magnitude I should expect from an unprotected relay coil at 12V, and how does a flyback diode + ferrite bead suppress it?"*

## 5 | BASIC ARCHITECTURE & CONNECTIONS

| Phase You Review/Own | Input From | Your Gate Decision Affects |
|---|---|---|
| Phase 2B Reviewer | M2: encoder ISR, correction layer, stall detection firmware | M2 cannot proceed to Phase 4 without your sign-off |
| Phase 3B Reviewer | M3: Kalman filter, optical flow, occlusion handling firmware | M3 cannot contribute to Phase 4 without your sign-off |
| Phase 6 Co-Owner | M4+M2+M3+M1+M6: full integrated system | Final 30-cycle acceptance test; all members witness |
| Phase 5 Electromagnet Check | M4 integration: relay log + encoder position during pick/drop | Detect magnetic interference on encoder lines |
| Phase 6 Kalman Guard | M3: P[0][0] and P[1][1] covariance monitor | Divergence must be caught; reset logged |
| Phase 6 Power Hardening | M5 owns: capacitor installation, ferrite bead, ground return | Zero encoder drift during relay events required |

**Your Gate Authority:** Every HARD GATE is your responsibility to document as PASSED. No team member advances without your written sign-off on the verification test log.

## 6 | TIMELINE & MILESTONES

**[ DAY 1 ]**

- Create master test log template: columns for Test ID, Date, Tester, Measured Value, Pass Threshold, Result
- Pre-read all verification tests: TEST 2B-A through 2B-E, TEST 3B-A through 3B-E
- Set up measurement rig: dial indicator, oscilloscope, calibration grid
- Install Python + pyserial + matplotlib for automated data capture

**[ PHASE 2B REVIEW (Days ~6–8) ]**

- TEST 2B-A: Command +90° on M1, confirm enc1_count = +600 counts (ENC_CPR/4)
- TEST 2B-B: Manually hold linkage during move; verify correction attempt triggers and logs
- TEST 2B-C: Block motor shaft for 600ms; verify FAULT state within 700ms
- TEST 2B-D: 10 pick positions; log encoder error each; all must be < ±1.5 deg
- TEST 2B-E: Idle 30 seconds; confirm encoder counts unchanged (zero drift)
- Document all results; countersign with M2; Phase 2B gate: PASS or BLOCKED

**[ PHASE 3B REVIEW (Days ~9–11) ]**

- TEST 3B-A: 50-frame stationary test; compute fused position stddev; must be < 1mm
- TEST 3B-B: Cover object for 3 frames; verify prediction within 5mm of truth
- TEST 3B-C: Move object at ~20mm/s; verify Vx/Vy sign and magnitude within 30%
- TEST 3B-D: Time full fusion pipeline (centroid + flow + Kalman); must be < 100ms
- TEST 3B-E: Remove object; verify no false transmit from noise
- Document all results; countersign with M3; Phase 3B gate: PASS or BLOCKED

**[ PHASE 6 — ROBUSTNESS & ACCEPTANCE (Days ~14–16) ]**

- Install power hardening: 1000µF across DRV8825 VMot, 100nF at VCC and encoder pins
- Run LED ring invariance check: TEST 3-B under LED ring only (ambient lights off)
- Verify Kalman divergence guard: force covariance spike; confirm reset logged

- Configure watchdog timers: 3-second WDT on both ESP32 and ESP32-CAM
- Configure 50ms stall timer: verify hardware timer triggers check_stall() correctly
- Run 30-cycle unattended acceptance test — all 6 members must witness
- Verify PASS 1–6 criteria; sign project sign-off sheet
- Log final Performance Specification v2 table with measured values

## 7 | ESCALATION MATRIX

| Situation | Action | Contact |
|---|---|---|
| TEST 2B-C: Stall takes > 700ms to detect | Log exact timing; FAIL the gate | M2 — reduce STALL timer accumulator threshold |
| TEST 3B-A: Fused stddev > 1mm | Log raw vs. fused comparison; FAIL the gate | M3 — increase R_MEAS; rerun test |
| TEST 3B-B: Occlusion recovery error > 5mm | Log prediction trajectory; FAIL the gate | M3 — check Q_VEL, verify optical flow valid flag |
| 30-cycle run: unrecoverable FAULT before cycle 27 | Stop run; document failure cycle and mode | M4 (integration) + M6 (systems) jointly |
| Encoder drift during relay switching (Phase 5/6) | Log drift magnitude and timing; install ferrite bead | M4 — confirm relay coil flyback suppression |
| Kalman divergence during 30-cycle run | Record cycle number and P values; confirm reset triggered | M3 — review divergence guard threshold |
| Any gate sign-off dispute with Phase Owner | Document disagreement with evidence | M6 — final arbitration authority |

***You are the standard-bearer for the whole team. Every PASS you sign carries the integrity of the entire system. Your rigor is what turns good code into a reliable machine. Hold the line, document everything, and celebrate every honest PASS.*** ■